

OpenClaw Navigation Transfer

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1. Course Content

Learning Objectives

- This section is a comprehensive course that combines robotic arm visual grasping and intelligent navigation functions for complex long-flow tasks. Please complete the basic courses [02-OpenClaw Robotic Arm Tracking Grasp], [05-OpenClaw Slam Mapping and Navigation], and [Road Network Planning Navigation] first.

2. Preparation

- Complete grid map construction, pose map construction, and road network file annotation. For details, refer to the tutorial [Road Network Planning Navigation - Building Pose Maps and Road Network Annotation]
- Complete the prerequisite tutorials, mastering map construction, road network annotation, and map mapping marking functions in advance.
- Start the MicroROS chassis agent (no need to restart if already started)

```
sh start_agent.sh
```

- Start odometry, tf, robotic arm assistance, camera nodes, etc. If you need to enable visual relocalization, add the startup parameter relocation:=True
- Start with relocalization (start from host)

```
ros2 launch m3pro_bringup car_base.launch.py relocation:=True
```

- Start road network navigation (start inside roadnet container)

```
ros2 launch road_net_route roadnet_nav2.launch.py  
rviz_config_file:=/root/volumes/code_ws/src/road_net_route/rviz/nav2_relocaton.rviz
```

- Start the MCP service

```
ros2 launch m3pro_bringup mcp_service.launch.py
```

- If you need voice reply, you need to start openclaw_bridge additionally, otherwise it can be omitted

```
ros2 run multi_brains_pre openclaw_bridge
```

3. Demonstration Case

[!TIP]

- You can freely edit commands based on actual conditions. The following is a reference case.
- **Transport a purple block from the raw material warehouse to the blue shelf area of the processing center**
- Use the robot_control CLI to pre-mark the map mapping for "raw material warehouse" and "processing center"

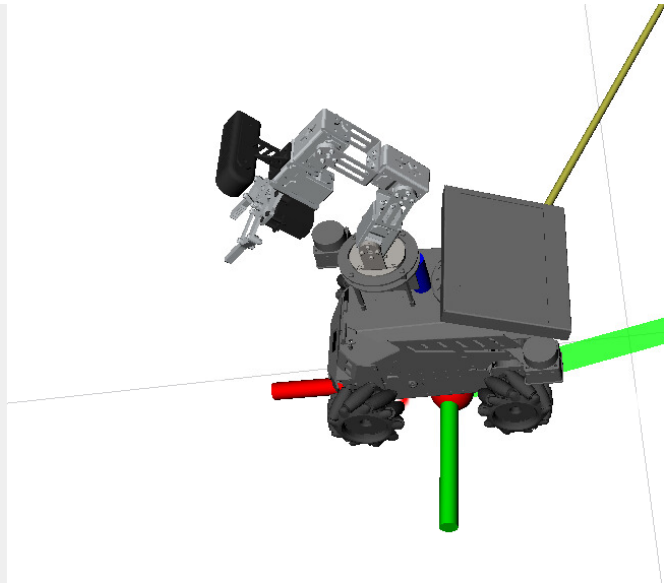
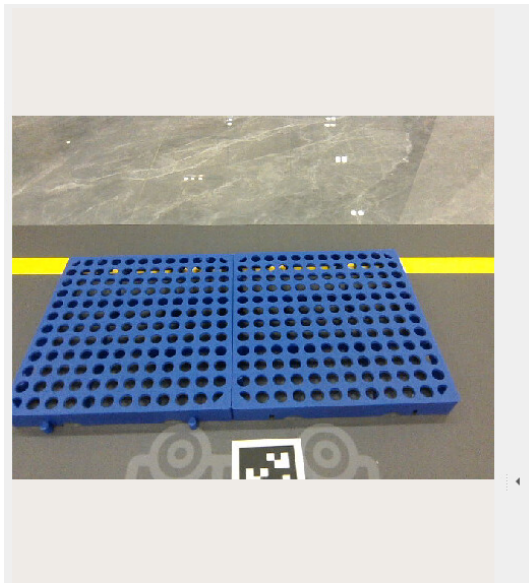
```
E:
name: 原料仓库
position:
  x: 0.155
  y: -2.047
  z: 0.0
orientation:
  x: 0.0
  y: 0.0
  z: -0.732
  w: 0.681
F:
name: 加工中心
position:
  x: 2.422
  y: -2.665
  z: 0.0
orientation:
  x: 0.0
  y: 0.0
  z: -0.707
  w: 0.707
```

- Then assign a specific task to OpenClaw, choose any interaction method
- Transport a purple cube from the raw material warehouse to the blue shelf area of the processing center.

```
connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.6-plus | tokens ?/1.0m
```

从原料仓库搬运一个紫色方块到加工中心蓝色货架区域

- OpenClaw will call tools to first grasp the color block and then navigate to the location for placement



4. Parameter Tuning and Source Code Analysis

- This section is a comprehensive course. Debugging parameters and functional source code are detailed in [02-OpenClaw Robotic Arm Tracking Grasp] and [05-OpenClaw Slam Mapping and Navigation].